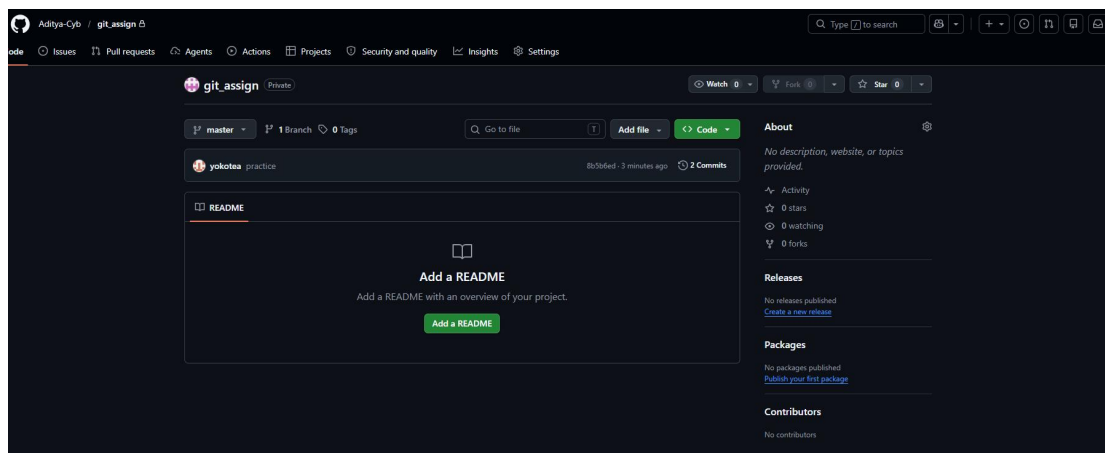


Assignment-4

1. Create a new GitHub repository.

- Clone the repository to your local machine using SSH (generate an SSH key if needed, add the public key to your GitHub account).
- Create a new branch named after your username (e.g., **Tutedude**).
- Add your Flask project files to this branch.
- Commit the changes and merge the branch into the **main** branch

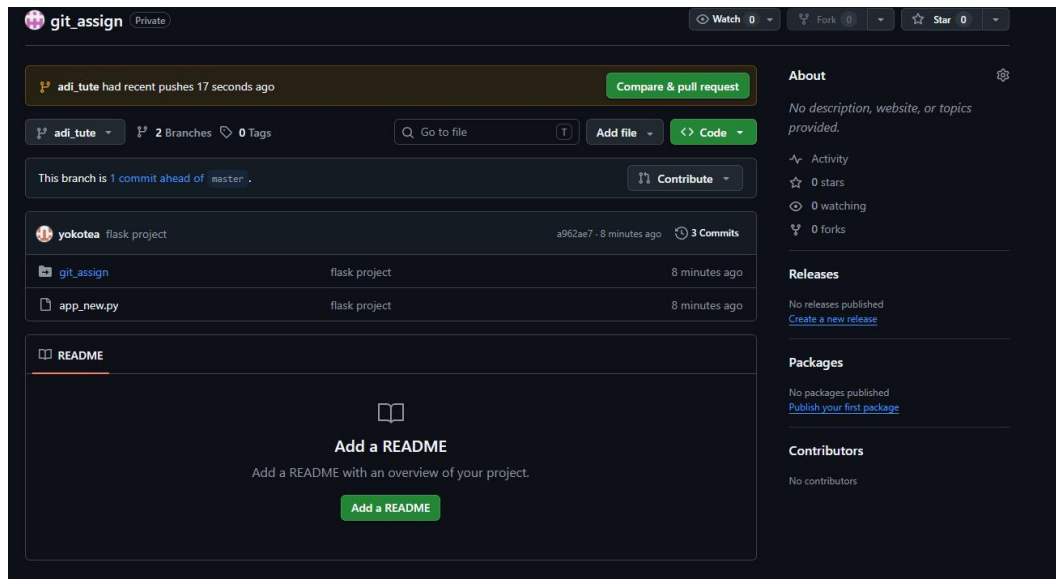


--> Repository Created named: git_assign

```
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/flask/assignment (master)
$ git clone git@github.com:Aditya-Cyb/git_assign.git
Cloning into 'git_assign'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), done.

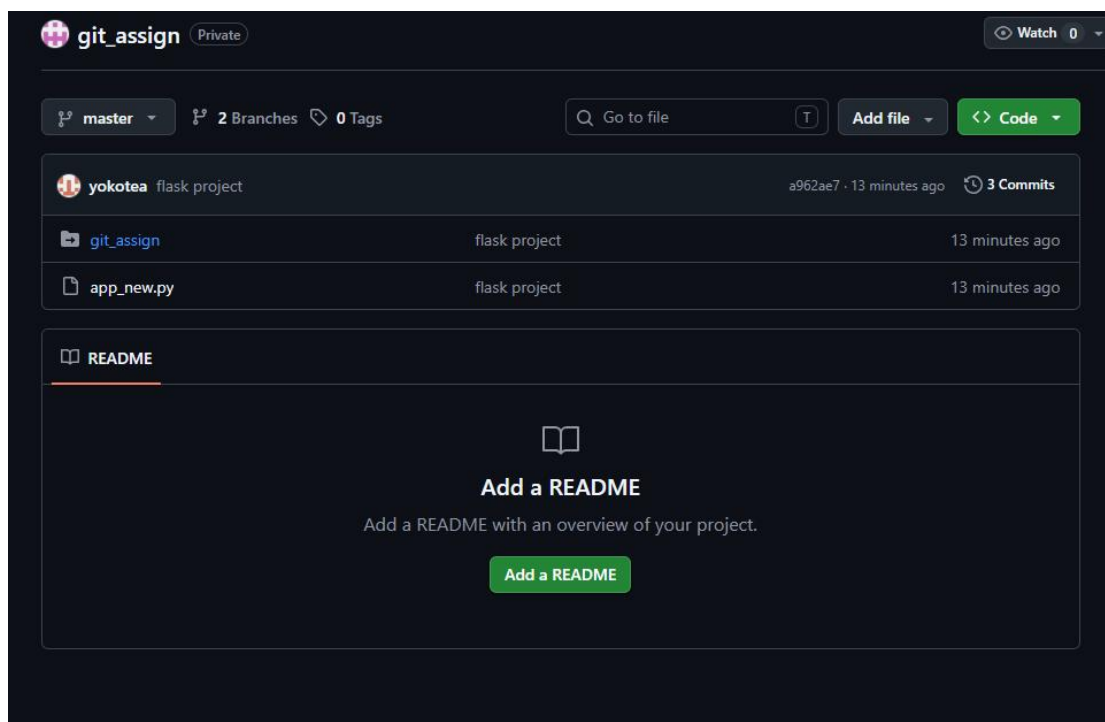
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/flask/assignment (master)
$ |
```

--> Repository Cloned



--> New branch - adi_tute

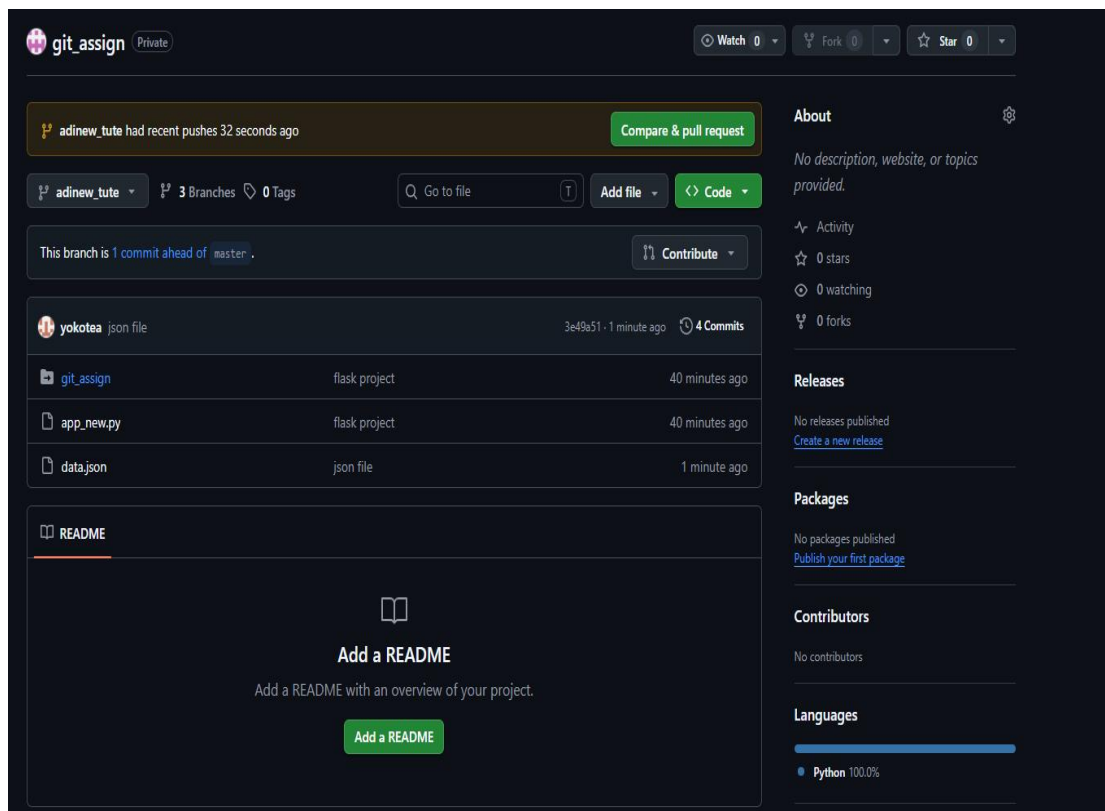
--> Porject added in the Branch- app_new.py



--> adi_tute branch merged with master granch

2. Create a new branch named `<your_name>_new` (e.g., `Tutedude_new`).

- Update the content of the JSON file used for the `/api` route in this branch.
- Merge the `<your_name>_new` branch into the `main` branch.
- If there are conflicts during the merge, resolve them by accepting the changes from the `<your_name>_new` branch.
- Add the resolved changes to the staging area, commit them, and push the updates to the remote repository.



--> New branch created: `adinew_tute`

```
git_assign / data.json in adinew_tute Cancel changes Commit changes

Edit Preview Spaces 2 No wrap

21 {
22   "id": 4,
23   "name": "Alice Brown",
24   "email": "alice@example.com",
25   "status": "active"
26 }
27 },
28 "products": [
29 {
30   "id": 101,
31   "name": "Laptop",
32   "price": 999.99,
33   "category": "Electronics"
34 },
35 {
36   "id": 102,
37   "name": "Mouse",
38   "price": 29.99,
39   "category": "Accessories"
40 },
41 {
42   "id": 103,
43   "name": "Keyboard",
44   "price": 79.99,
45   "category": "Accessories"
46 },
47 {
48   "id": 104,
49   "name": "Phone",
50   "price": 9999.99,
51   "category": "Device"
52 }
53 ]
54 }
55 }
```

Use **Control + Shift + ↵** to toggle the **tab** key moving focus. Alternatively, use **esc**, then **tab** to move to the next interactive element on the page.

--> data.json file updated

git_assign Private

Watch 0 Fork 0 Star 0

adinew_tute had recent pushes 2 minutes ago [Compare & pull request](#)

master 3 Branches 0 Tags

Go to file

Add file <> Code

yokotea json file 3e49a51 · 11 minutes ago 4 Commits

git_assign flask project 51 minutes ago

app_new.py flask project 51 minutes ago

data.json json file 11 minutes ago

README

Add a README

Add a README with an overview of your project.

Add a README

About

No description, website, or topics provided.

Activity 0 stars 0 watching 0 forks

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Contributors

No contributors

Languages

Python 100.0%

Suggested workflows

--> Branch - adinew_tute merged with the main branch - master without any error.

3. Branch Creation:

Create two branches: **master_1** and **master_2** from the **main** branch.

Feature Development in **master_1**:

In the **master_1** branch, create a **To-Do Page** in the frontend.

The page should contain a form with the following fields:

Item Name

Item Description

Backend API in **master_2**:

In the **master_2** branch, create a backend route named **/submittodoitem**.

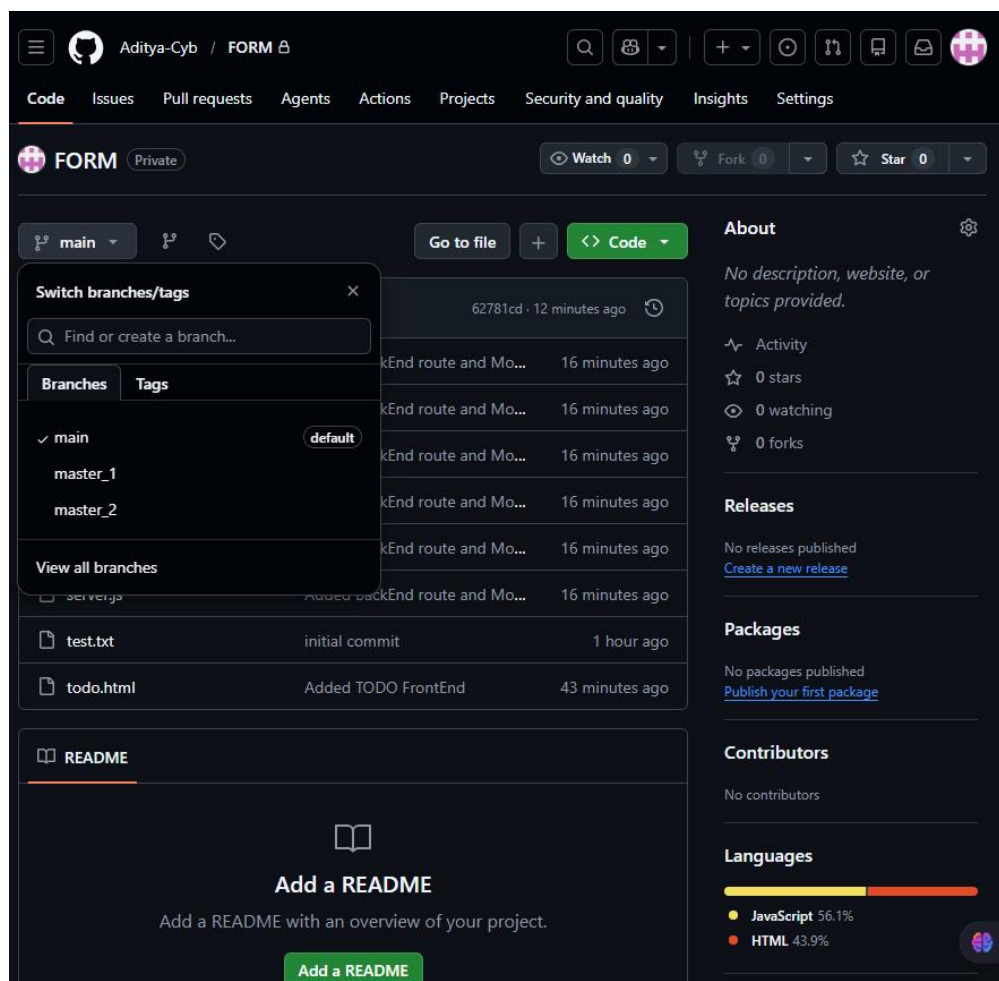
This route will:

Accept **itemName** and **itemDescription** via a POST request.

Store these details in a MongoDB database.

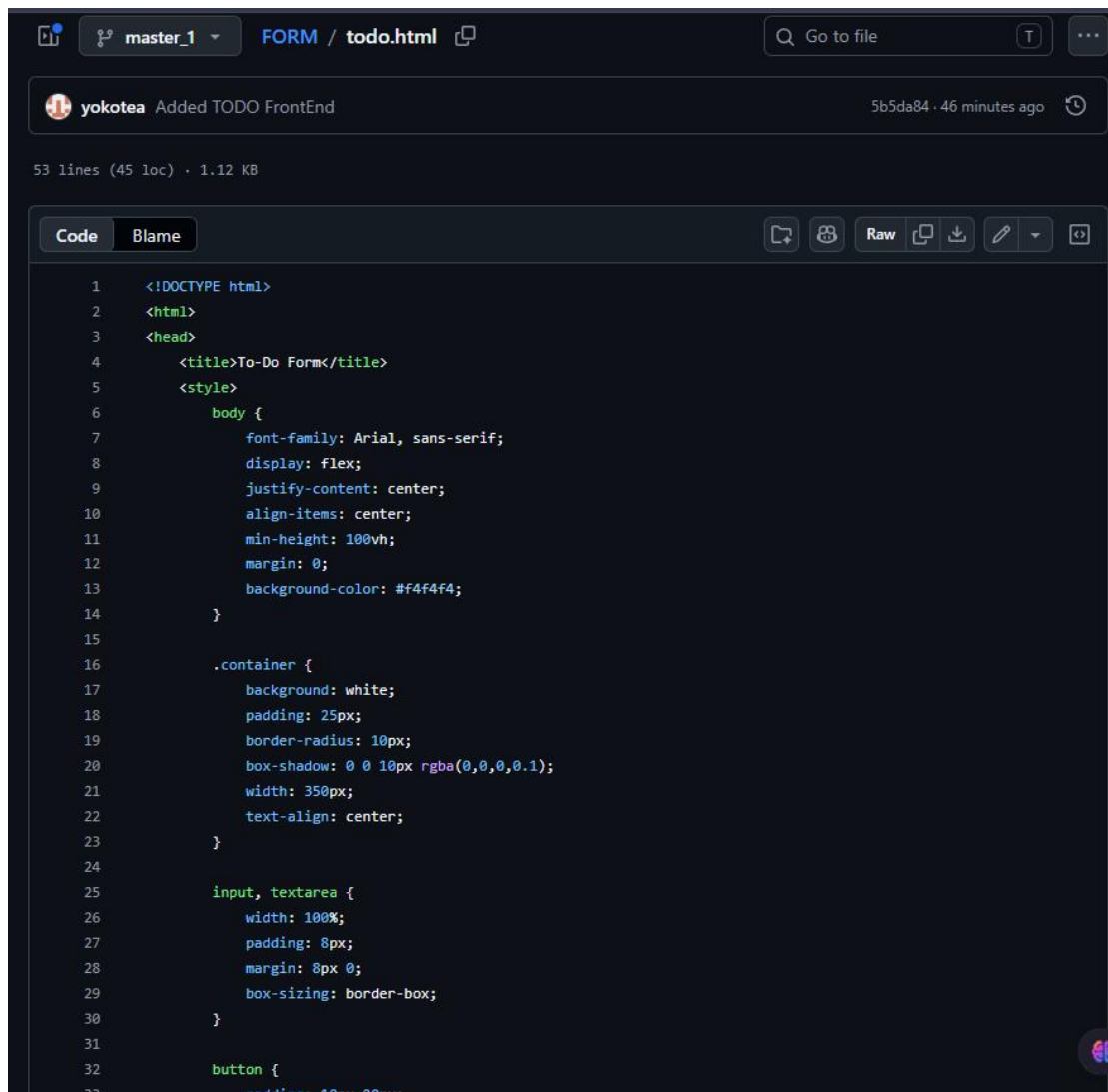
Merging Changes:

Merge the changes from both **master_1** and **master_2** into the **main** branch.



--> Two Branches : master_1 and master_2 created

--> **master_1**



The screenshot shows a code editor interface with a dark theme. At the top, the breadcrumb navigation indicates the current file is `FORM / todo.html` in the `master_1` branch. A commit message "yokotea Added TODO FrontEnd" is visible, along with the commit hash `5b5da84` and the time "46 minutes ago". Below this, the file statistics show "53 lines (45 loc) · 1.12 KB". The editor has tabs for "Code" and "Blame", with "Code" being the active tab. The code is HTML with embedded CSS. The CSS defines a `body` with a sans-serif font, flex display, and a light blue background. A `.container` class is used for the main content area, featuring a white background, padding, a border-radius, and a box-shadow. Input and text area elements are styled with 100% width and 8px padding. A `button` class is also defined with 10px padding.

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>To-Do Form</title>
5   <style>
6     body {
7       font-family: Arial, sans-serif;
8       display: flex;
9       justify-content: center;
10      align-items: center;
11      min-height: 100vh;
12      margin: 0;
13      background-color: #f4f4f4;
14    }
15
16    .container {
17      background: white;
18      padding: 25px;
19      border-radius: 10px;
20      box-shadow: 0 0 10px rgba(0,0,0,0.1);
21      width: 350px;
22      text-align: center;
23    }
24
25    input, textarea {
26      width: 100%;
27      padding: 8px;
28      margin: 8px 0;
29      box-sizing: border-box;
30    }
31
32    button {
33      padding: 10px 20px;
```

FrontEnd Created : todo.html

-->master_2

The screenshot displays the GitHub interface for a repository named 'FORM' by user 'Aditya-Cyb'. The repository is private and currently has 0 forks and 0 stars. The selected branch is 'master_2', which is 2 commits behind the 'main' branch. A commit by 'yokotea' is shown, titled 'Added BackEnd route and MongoDB integrat...', committed 21 minutes ago. Below the commit, a table lists the files added in this commit:

File	Commit Message	Time
models	Added BackEnd route and Mo...	21 minutes ago
node_modules	Added BackEnd route and Mo...	21 minutes ago
routes	Added BackEnd route and Mo...	21 minutes ago
package-lock.json	Added BackEnd route and Mo...	21 minutes ago
package.json	Added BackEnd route and Mo...	21 minutes ago
server.js	Added BackEnd route and Mo...	21 minutes ago
test.txt	initial commit	1 hour ago

Below the file list, there is a section for 'README' with a button to 'Add a README'. The sidebar on the right contains sections for 'About' (no description), 'Activity' (0 stars, 0 watching, 0 forks), 'Releases' (no releases published), 'Packages' (no packages published), 'Contributors' (no contributors), and 'Languages' (JavaScript 56.1%, HTML 43.9%).

BackEnd Created

--> merge of master_1 and master_2 into Main branch

4. Enhancing the To-Do Form in `master_1`:

In the `master_1` branch, add the following fields to the To-Do form:

Item ID

Item UUID

Item Hash

Committing in Sequence:

Add and commit each field separately in the following order:

First commit: Add **Item ID** field.

Second commit: Add **Item UUID** field.

Third commit: Add **Item Hash** field.

Merging to `main`:

Merge the `master_1` branch into the `main` branch.

Git Reset and Commit Deletion:

In the `main` branch, use **Git Reset** to roll back to the commit where only the **Item ID** field was added.

Use `git reset --soft` to ensure changes remain staged.

Re-commit this state to the `main` branch.

Merge this updated state to the `main` branch

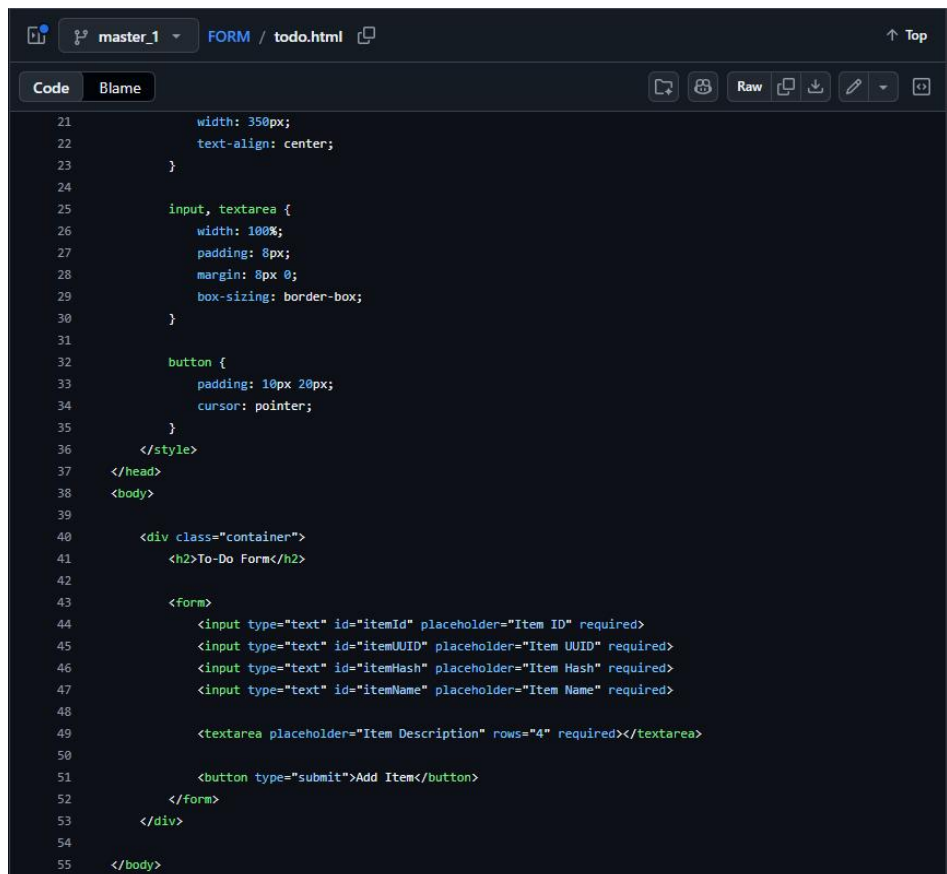
Rebasing Changes:

Rebase the updated changes in the `main` branch to the `master_1` branch.

Clarification:

During rebasing, **preserve individual commits** to maintain the commit history for each change (i.e., do not squash commits).

Use `git rebase main master_1` to integrate changes from the `main` branch back into the `master_1` branch.



```
21     width: 350px;
22     text-align: center;
23   }
24
25   input, textarea {
26     width: 100%;
27     padding: 8px;
28     margin: 8px 0;
29     box-sizing: border-box;
30   }
31
32   button {
33     padding: 10px 20px;
34     cursor: pointer;
35   }
36 </style>
37 </head>
38 <body>
39
40   <div class="container">
41     <h2>To-Do Form</h2>
42
43     <form>
44       <input type="text" id="itemId" placeholder="Item ID" required>
45       <input type="text" id="itemUUID" placeholder="Item UUID" required>
46       <input type="text" id="itemHash" placeholder="Item Hash" required>
47       <input type="text" id="itemName" placeholder="Item Name" required>
48
49       <textarea placeholder="Item Description" rows="4" required></textarea>
50
51       <button type="submit">Add Item</button>
52     </form>
53   </div>
54
55 </body>
```

Field(ID , UUID , Hash) added in branch master_1

```
MINGW64:/c:/Users/LENOVO/Desktop/TODO
Author: aditya dutta <aditya123@gmail.com>
Date: Mon Jun 15 20:48:15 2026 +0530

Merge branch 'master_1'

commit a280149aabf293221ae4508556d86fec5441d9e2 (origin/master_1, master_1)
Author: aditya dutta <aditya123@gmail.com>
Date: Mon Jun 15 20:47:39 2026 +0530

Added Item Hash Field

commit 0695100b0a37a9c65dc5f264778866a7a4cd51f2
Author: aditya dutta <aditya123@gmail.com>
Date: Mon Jun 15 20:46:51 2026 +0530

Added Item UUID Field

commit f520095f0e5833d48854a232bf3fd43676c58486
Author: aditya dutta <aditya123@gmail.com>
Date: Mon Jun 15 20:45:46 2026 +0530

Added Item ID field
```

Commit done in a required sequence

```
main FORM / todo.html
Code Blame
21 width: 350px;
22 text-align: center;
23 }
24
25 input, textarea {
26 width: 100%;
27 padding: 8px;
28 margin: 8px 0;
29 box-sizing: border-box;
30 }
31
32 button {
33 padding: 10px 20px;
34 cursor: pointer;
35 }
36 </style>
37 </head>
38 <body>
39
40 <div class="container">
41 <h2>To-Do Form</h2>
42
43 <form>
44 <input type="text" id="itemId" placeholder="Item ID" required>
45 <input type="text" id="itemUUID" placeholder="Item UUID" required>
46 <input type="text" id="itemHash" placeholder="Item Hash" required>
47 <input type="text" id="itemName" placeholder="Item Name" required>
48
49 <textarea placeholder="Item Description" rows="4" required></textarea>
50
51 <button type="submit">Add Item</button>
52 </form>
53 </div>
54
55 </body>
56 </html>
```

Merge of master_1 completed with main branch

```
MINGW64:/c/Users/LENOVO/Desktop/TODO
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (main)
$ git log --oneline --all
e21efd4 (origin/main) Merge branch 'master_1'
a280149 (origin/master_1, master_1) Added Item Hash Field
0695100 Added Item UUID Field
f520095 (HEAD -> main) Added Item ID field
62781cd Merge branch 'master_2'
2be79db (origin/master_2, master_2) Added BackEnd route and MongoDB integration
5b5da84 Added TODO FrontEnd
3a9d9d8 initial commit

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (main)
$ git commit -m "Recommitted ID state"
[main b7f89bd] Recommitted ID state
```

```
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (main)
$ git log --oneline --all
b7f89bd (HEAD -> main) Recommitted ID state
e21efd4 (origin/main) Merge branch 'master_1'
a280149 (origin/master_1, master_1) Added Item Hash Field
0695100 Added Item UUID Field
f520095 Added Item ID field
62781cd Merge branch 'master_2'
2be79db (origin/master_2, master_2) Added BackEnd route and MongoDB integration
5b5da84 Added TODO FrontEnd
3a9d9d8 initial commit
```

In the main branch, used:

```
git reset --soft <Item_ID_commit_hash>
```

to roll back to the commit where only the Item ID field was added while keeping changes staged.

Recommitted the reset state in the main branch.

```
MINGW64:/c/Users/LENOVO/Desktop/TODO

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (main)
$ git rebase main master_1
Auto-merging todo.html
CONFLICT (content): Merge conflict in todo.html
error: could not apply 0695100... Added Item UUID Field
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --
abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 0695100... # Added Item UUID Field

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (master_1|REBASE 1/2)
$ git log --oneline --graph --all
* b7f89bd (HEAD, main) Recommitted ID state
| * e21efd4 (origin/main) Merge branch 'master_1'
|/
| * a280149 (origin/master_1, master_1) Added Item Hash Field
| * 0695100 Added Item UUID Field
|/
| * f520095 Added Item ID field
| * 62781cd Merge branch 'master_2'
|/
| * 2be79db (origin/master_2, master_2) Added BackEnd route and MongoDB integration
| * 5b5da84 Added TODO FrontEnd
|/
* 3a9d9d8 initial commit

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (master_1|REBASE 1/2)
$ git add todo.html

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (master_1|REBASE 1/2)
$ git rebase --continue
dropping a280149aabf293221ae4508556d86fec5441d9e2 Added Item Hash Field -- patch contents alr
eady upstream
Successfully rebased and updated refs/heads/master_1.

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/TODO (master_1)
$ |
```

Rebased master_1 onto main to synchronize branch histories. Merge conflicts were resolved manually, and the rebase was completed successfully using git rebase --continue